Hooke’s law

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Learning target: Connect spring displacement with restoring force.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 11; 3 course days later.*
3 J of rotational kinetic energy. A rolling wheel also spins.

02. Explain a worked example (worked)
Displacement away from equilibrium; force points back, with 3 N magnitude.

03. Look and explain (model)
Stretched right gives force left; compressed left gives force right.

04. Build the solution (practice)
3 N and 6 N.

05. Use the idea (reason)
Force doubles; k is constant within linear range.

06. Choose a method (mixed) *Review: Unit 4, Day 2; 12 course days later.*
+35 J; force and displacement point the same way.

07. Exit check (exit)
8 N back toward natural length.

08. STRETCH: Apply it another way (stretch)
0.08 and 0.04 m; higher k needs less stretch for the same force.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 11; 3 course days later.*
$v=\sqrt{(2\ gh)}\approx 3.464$ m/s.

02. Explain a worked example (worked)
Displacement away from equilibrium; force points back, with 3 N magnitude.

03. Represent and interpret (model)
Stretched right gives force left; compressed left gives force right.

04. Solve and justify (practice)
k 196 N/m; force 23.52 N.

05. Analyze the reasoning (reason)
Force doubles; k is constant within linear range.

06. Choose a method (mixed) *Review: Unit 4, Day 2; 12 course days later.*
F=6 N; zero displacement gives 0 J.

07. Exit check (exit)
+12 N, right.

08. STRETCH: Apply it another way (stretch)
k 150 N/m; F=+6 N. Sign encodes restoring direction.